behaving, just as we have with cameras, cars, telephones and personal computers. We will need to change some laws and establish new institutions. It will be uncomfortable for everyone but it is a surmountable problem.

That broad topic bears on this e-book too. The intelligence agencies, like everyone else, are grappling with new capabilities, vulnerabilities and constraints. The American constitution, with its admirable prohibition of arbitrary searches and seizures, was not designed for the digital age, which allows large amounts of information to be stored, and sorted rapidly and repeatedly by many remote investigators. The *Smith v Maryland* Supreme Court judgment of 1979, which says that expectations of privacy do not apply to dialled telephone numbers, did not foresee the ability of the authorities to warehouse such material in colossal amounts and search it automatically.

In sufficient quantities and in the right context, meta-data (for example the time, duration and direction of a phone call) can be more revealing than its content. The latter requires a search warrant. The former does not. Similarly the American distinction between citizens (constitutionally protected) and foreigners (fair game) is impractical and outdated when data crosses and re-crosses national frontiers. The need to rely on (and to arm-twist) private internet, technology and telephone companies creates new problems and dilemmas. In particular, weakening commercial cryptography standards, or finding vulnerabilities and keeping them secret, for example, may bring immediate gains in terms of intelligence collection, but—if exposed—strategic losses in terms of stoking distrust of America as a guardian of global security standards.

Legal challenges on these issues in America are under way as I write this e-book and will doubtless reach the Supreme Court eventually. Political controversies are bubbling too. The presidential Review Group on